

De novo genome assembly and analyses of 12 founder inbred lines provide insights into maize heterosis

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Supplementary Information

Supplementary Note:

Additional Methods:

Phylogenetic analyses

The Illumina paired-end reads of 12 FILs were mapped to the B73_v3 reference genome (RefGen_v3 using BWA mem (v.0.7.13) with default parameters, following the mapping processing by SAMtools and SNP calling using GATK (v.3.5-0-g36282e4, with best-practice workflows) as previously described¹. The resulting SNPs were compared with the reported 25 million SNPs in 350 inbred lines (based on B73_v3 genome)¹. 13,723,928 overlapped SNPs were retained for phylogenetic analyses. A model based approach implemented in the ADMIXTURE tool (v.1.3.0)² was used to estimate the ancestry population number (K) for these 362 inbred lines (350 maize inbred lines plus 12 FILs). We noticed that when K=4, the cross-validation error reached a turning point, consistent with the earlier report¹. While when K=10, the cross-validation error reached the lowest, hinting that the 362 inbred lines might be originated from 10 ancestries. The maximum-likelihood tree (ML-tree) was constructed using the FastTree (v.2.1.10) software with the default parameters³.

RNA-seq data collection for genome annotation

To assist genome annotation, we generated two set of RNA-seq data for the 12 FILs. The first one includes an equal mix of RNAs from 8 tissues that consist of seedlings in V2 stage (with 2 expanded leaves), husk, tassel, cob, ear leaf, leaf sheath, stem, and aerial root of the VT stage (the tasseling stage). The second set includes 5 tissues that consist of VE (the stage when the first leaf emerged but not expanded yet) shoot, VE root, ovule before pollination, kernels of 5 and 10 days after pollination. Samples were collected in the greenhouse (seedlings) or field of Langfang city (116° 70'E, 39° 52'N) in Hebei province in 2018. Total RNA for individual sample was extracted using TRIzol reagent (Invitrogen). The RNA concentrations were measured using NanoDrop 2000C. Equal amount of RNA from different samples were mixed as described above to form two RNA sets for library construction. The libraries were sequenced by paired-end sequencing strategy using Illumina HiSeq 4000 platform.

We generated >10 Gb raw data for each RNA set (and thus resulting in >20 Gb RNA-seq data for each FIL, Supplementary Table 2).

Repetitive sequences analysis

The repetitive sequences in all 12 newly assembled maize genomes, combined with B73 (B73_v4) and Mo17 genomes were annotated using a comprehensive transposable annotation pipeline EDTA⁴ (<https://github.com/oushujun/EDTA>, v1.7.1). Briefly, each genome was annotated using the parameters of -genome genome_sequence -species others -step all -overwrite 1 -sensitive 0 -anno 1 -threads 20. The curated TE sequence library was stored in the file of TElib.fa, and the content of each TE class was extracted from the file of TEanno.sum.

To calculate the percentage of TE in the nonsyntenic regions, we first used bedtools⁵ (v2-2.25.0) to remove the syntenic blocks on each chromosome, yielding the nonsyntenic blocks (command: bedtools subtract -a genome.bed -b sys.bed); Then the overlapping of each nonsyntenic block with TE was calculated (command: bedtools intersect -b nonsys.bed -a TE.bed -wo). Finally, we calculated the total length of TEs and total nonsyntenic length, as well as the total length of N in the nonsyntenic regions, then to calculate the percentage of TE by dividing the total length of TEs by the total length of non-N sequences in the nonsyntenic regions. The percentage of TE in the syntenic regions was calculated using a similar procedure: bedtools was used to calculate the overlapping of TE with the syntenic blocks, then the percentage of TE was calculated by dividing the total length of overlapped TEs by the total length of non-N sequences in the syntenic regions.

Gene functional annotation

Five databases: GO_Annotation (by blast2go⁶ v2.5, default setting), KEGG_Annotation (by blastp v2.2.31, with parameters: -num_descriptions 100 -num_alignments 100 -evaluate 1e-5), KOG_Annotation (by blastp v2.2.31, with parameters: -num_descriptions 100 -num_alignments 100 -evaluate 1e-5), TrEMBL_Annotation (by blastp v2.2.31, with parameters: -num_descriptions 100 -num_alignments 100 -evaluate 1e-5) and NR_Annotation (by blastp v2.2.31, -max_target_seqs 100 -evaluate 1e-5) were used for gene functional annotation (Supplementary Table 9).

Identification of PAV sequences and PAV genes

PAV sequences were identified using a sliding-window method as described⁷. Briefly, the genome of a FIL (Query) was divided into 500 bp overlapping windows with a step size of 100 bp and each 500 bp window (without N or n) was aligned respectively against the rest 13 FILs genome (References) with BWA (v0.7.13) mem⁸ with options ‘-w 500 --M’. A total of 182 (14×13) alignments were performed among the 14 FILs. The sequences for windows that could not be aligned or with aligned coverage less than 25% to the Reference genome were defined as specific sequences for the Query genome. The overlapped windows were merged. To verify the reliability of these PAVs, high depth Illumina sequencing data (~20×) of the Query FIL was mapped to the Reference genome using BWA (v0.7.13) mem⁸. The putative PAVs with less than 50% sequences covered (calculated by samtools 1.11) by reads from Query FIL were defined as the true Reference-specific sequences, and vice versa. Subsequently, genes with more than 75% CDS sequences covered by PAV sequences were defined as PAV genes. The reported resequencing data of 19 wild relatives, 23 landraces from maize Hapmap2⁹ were mapped to the 14 FIL genomes using BWA (v.0.7.13) mem⁸ with the default parameters to test if the detected PAVs were pre-existed in the wild relatives of maize. For Gene Ontology enrichment analysis of the PAV genes, the PAV and background genes were annotated using PANNZER2¹⁰ followed by an enrichment test ($p < 0.05$) using Fisher exact test of the fisher package in Python 3.7.6.

QTL analysis of MRDD

A recombinant inbred lines (RILs) population with 670 lines derived from a cross of Xu178×HuangC was used for QTL analysis¹¹. The traits of MRDD were investigated in Kaifeng City (Henan Province, China) in the summer of 2018 under natural conditions. The susceptibility of MRDD was phenotyped and scored as reported¹². QTL mapping was performed using the composite interval mapping method in the R/qtl package (v.1.48-1). The cutoff and confidence intervals for QTL identification were defined by 1,000 permutations ($p < 0.05$) approach and 1.5 LOD-drop method.

RNA-seq data collection for eQTL analysis

Immature ears at the silking stage (before pollination, the ears from 3 independent plants were mixed to form a sample) were collected for RNA extraction and

sequencing as described above. Totally, 131 inbred lines, a subset of the 350 maize panel¹, were planted in Langfang City in Hebei province of China in the summer of 2018, for RNA-seq analysis. Approximately 6 Gb Illumina raw data were generated for each IL. After clipping the adaptor sequences and removing the low quality reads, the RNA sequence data for each sample were mapped to the 14 assemblies independently using Hisat2-2.0.4¹³ with the default parameters. The bam files of uniquely mapped reads were used as inputs for the Cufflinks (v2.2.0) software¹⁴, and per kilobase of transcript per million mapped reads (FPKM) were calculated to measure the expression levels of genes using the DESeq2 Bioconductor package¹⁵.

eQTL analysis

The eQTL analysis was carried out using R package “MatrixeQTL” based on the “modelLINEAR” model. The local eQTL mapping window was set as 33 kb (the genome-wide averaged distance of LD decayed to $r^2 = 0.2$ in the 350 maize panel¹). A cutoff of p -value $< 1e-6$ (Benjamini–Hochberg FDR < 0.05) was used for defining significant signals.

For candidate association mapping of *ZmLOX3*, Zm00001d006055 and Zm00001d011140 expression, the genotypes of SVs and the identified SNPs¹ were combined as the genotype data, and the expression level (FPKM value) of *ZmLOX3*, Zm00001d006055 and Zm00001d011140 in 131 inbred lines were treated as the phenotype data. The candidate association analysis was conducted using the MLM method in Tassel5 (v.5.2.22)¹⁶.

Additional Results:

Repetitive DNA sequence annotation

Repetitive elements are major components of maize genomes that play critical roles in regulating genome evolution and adaptation¹⁷. We used the extensive de-novo TE annotator (EDTA) pipeline⁴ to annotate the repetitive DNA in the 12 FIL genomes. In general, all 12 FILs showed highly similar transposable element (TEs) compositions, *i.e.* about 83.94-85.56% (an average of 84.68%) of their genomes were repetitive sequences (Supplementary Table 5). 11.92-17.16% (an average of 14.28%) were DNA transposons, 66.07-71.46% (an average of 69.54%) were long terminal repeat (LTR) retrotransposons, and 0.78-1.07% (an average of 0.86%) were miniature inverted-repeat transposable elements (MITEs). Among the DNA transposons, the

Helitron transposon was most abundant, with an average of 3,087 intact *Helitron* transposons (ranging from 2,925 in Dan340 to 3,174 in PH207). Among the retrotransposons, the *Gypsy* (ranging from 37.51-41.44%) and *Copia* (ranging from 16.36-19.81%) superfamilies were the most abundant (Supplementary Table 5). The mean number of intact *Gypsy* retrotransposons in the 12 FILs was 22,515 (ranging from 19,744 in Dan340 to 24,619 in Jing92), with an average accumulative length of 231,468,733 bp (ranging from 198,082,178 bp in Dan340 to 256,231,800 bp in Jing92). The mean number of intact *Copia* retrotransposons in the 12 FILs was 18,458 (ranging from 17,014 in Dan340 to 19,326 in Jing92), with an average accumulative length of 170,403,051 bp (ranging from 156,852,794 bp in Dan340 to 178,650,773 bp in Jing92).

We subsequently analyzed the distribution of *Gypsy*, *Copia* and *Helitron* TEs. Interestingly, we found that the *Gypsy* retrotransposons were enriched in the pericentromeric regions, whereas the *Copia* retrotransposons were enriched in the chromosome arms (Extended Data Fig. 1), similar to the reports for B73 and four European maize genomes^{18,19}. The distribution of *Helitron* transposons was similar to *Copia* and was depleted in the pericentromeric regions (Extended Data Fig. 1). In general, the compositions and numbers of TE families were similar among the 12 FILs and comparable to the previously reported B73_v4, Mo17 (Supplementary Table 5 and Extended Data Fig. 1) and 26 NAM founder genomes²⁰, indicating that no significant burst of TEs occurred during temperate maize breeding.

We next assessed the relationship of TE and gene distribution (Supplementary Table 10). For the three most abundant TEs, the distribution of genes was negatively correlated with that of *Gypsy* retrotransposons, but positively correlated with that of *Copia* and *Helitron* (Extended Data Fig. 1). Moreover, we found that among the various TEs, the LTR retrotransposons were always located far away from genes (an average of 28,805 bp and 45,054 bp away from genes for *Copia* and *Gypsy* retrotransposons respectively), whereas MITE TEs (an average of, an average of) and DNA TEs were preferentially located around genes. MITE TEs were on average 14,172 bp away from genes, 17.79% located in genic regions, and 2.64% in exons, while DNA TEs were on average 19,201 bp away from genes, 15.17% located in genic regions and 4.09% in exons (Supplementary Table 10). Among the three most abundant TE superfamilies, the *Helitron* TEs were located closest to genes and more frequently located in exons, consistent with the earlier report that *Helitron*-like transposons contributed to gene duplication, exon shuffling and thus genome evolution and

diversification⁴⁸. Among the other TE superfamilies, more than 50% of DTT TEs (Tc1-Mariner superfamily, including DNA/DTT and MITE/DTT) and DTH TEs (PIF-Harbinger superfamily, including DNA/DTH and MITE/DTH) were located near genes (~2kb regions). Notably, an average of 20.1% of DNA/DTH (18.0%-22.9%), 23.1% of DNA/DTT TEs (17.1%-28.2%), 16.8% of MITE/DTH TEs (15.1%-17.7%) and 32.5% of MITE/DTT TEs (29.4%-37.1%) were located in gene bodies. Moreover, an average of 4.7% of DNA/DTH (3.6%-5.1%), 4.4% of DNA/DTT TEs (3.3%-5.4%), 3.4% of MITE/DTH TEs (2.5%-3.7%), and 3.1% of MITE/DTT TEs (2.6%-3.7%) were located in exons. It was reported that *As-Ds* TEs (belong to DTA superfamily) prefer to insert into exons and this was used for maize mutant library construction⁴⁹. Our results showed that an average of 3.4% of DNA/DTA (2.6%-3.9%) and 2.2% of MITE/DTA (1.5%-2.5%) TEs were located in exons, which is less frequent than DTH and DTT TEs. These results suggest that the various TEs have distinct characteristics of distribution and transposition patterns, and their activities collectively contributed to the evolution and diversification of the maize genome.

WGA analysis of various heterotic groups

To further analyze the syntenic relationship of the 14 FILs, we mapped all the one-to-one syntenic blocks of the other 13 FILs to the genome of B73_v4, forming a pan-WGA-map based on B73. Strikingly, we found that only ~390 Mb (approximately 18.3% of B73 genome) were syntenic regions shared by all 14 FILs (Extended Data Fig. 2c-2d), reflecting extensive genetic variation within these FILs. Notably, we identified several genes involved in domestication or post-domestication spread of maize in these conserved regions, including *TGA1*²¹, *TBI*²², and *ZmCCT10*^{23,24}.

The WGA analysis also provided insight into the formation of various heterotic groups. As expected, the subtropical FIL S37 had the lowest percentage of synteny (an average of 51.90% WGA synteny) and Identity-By-State (IBS, represented by the syntenic blocks with >99.5% identity, an average of 27.12%) with the 13 temperate FILs, suggesting that it is more distantly related to the various temperate FILs (Fig. 3a). Among the temperate FILs, A632 and B73 are both SS-type germplasm and they shared a high degree of synteny (67.59% WGA synteny, 49.49% IBS). Zheng58 and Ye478 have close kinship and both belong to the PA group. Zheng58 was likely selected from the progeny of Ye478 crossed with an unknown parental line (named Inbred-X) likely derived from Dan340^{25,26}. Using a more stringent IBS analysis (more

than 99.9% identity), we found that 43.78% and 27.82% of the Zheng58 genome was derived from Ye478 and Dan340, respectively, and Dan340 accounts for ~49.49% of the non-Ye478 derived sequence in the Zheng58 genome, thus verifying that Ye478 and Dan340 are the major contributors to the Zheng58 genome (Supplementary Table 12). Both Ye478 and Zheng58 in general also had relatively high synteny (an average of 60.72% WGA synteny, 40.76% IBS) with A632 and B73, supporting the notion that the PA-group is likely derived from the SS-type germplasm. Our analyses also revealed that Dan340 shared a high degree of synteny (62.98% WGA synteny, 48.65% IBS) with OH43, hinting that Dan340 might share a common ancestor with OH43. The X-group germplasm represents a new heterotic group in maize breeding in China²⁷. Interestingly, we found that Jing724 has a closer relationship with PH207 (62.75% WGA synteny, 43.59% IBS), B73 and A632 (both belong to the SS-type) (an average of 61.21% WGA synteny, an average of 39.97% IBS), but was more distant from the TSPT-type FILs (Huangzaosi, Chang7-2 and Jing92) (an average of 53.48% WGA synteny, an average of 27.09% IBS), suggesting that Jing724 might be derived from the progeny of SS crossing with IDT. Xu178 is a FIL for the PB-group and is derived from the US hybrid 78599. Interestingly, it also had relatively low synteny with the other 12 temperate FILs (an average of 54.61% WGA synteny, an average of 32.06% IBS), suggesting that Xu178 is indeed genetically distinct from other temperate FILs and that it has the potential to form viable heterotic patterns with other Chinese heterotic groups. Also notably, the three TSPT-type FILs (Huangzaosi, Chang7-2 and Jing92) generally had low syntenic lengths with other FILs (an average of 53.59% WGA synteny, an average of 27.73% IBS) (Fig. 3a), reflecting its unique genomic composition and characteristics. The TSPT group is known to form excellent heterotic patterns with all other heterotic groups (particularly with PA- and X- groups)²⁸. In addition, Huangzaosi, Chang7-2 and Jing92 are known to have close kinship. Chang7-2 was directly derived from Huangzaosi, and Jing92 was derived from Chang7-2 (Supplementary Fig. 1). The ratios of one-to-one syntenic block-occupied genomes were high (69.22-72.74%) among Huangzaosi, Chang7-2 and Jing92. IBS analysis indicated that 57.78% of the Chang7-2 genome was directly inherited from Huangzaosi, whereas 25.43% and 23.53% of the Jing92 genome was directly inherited from Chang7-2 and Huangzaosi, respectively (Supplementary Table 11).

FIL specific sequences analysis

We aligned the resequencing data of 19 wild relatives and 23 landraces from maize Hapmap2 to the 14 FIL genomes. The results showed that an average of 93.4% of closely related homologs of the FIL specific sequences could be detected in at least one of the teosinte or landrace accessions (Supplementary Table 15), indicating that most of these PAVs are variants present in the wild relatives of maize, in line with a previous report⁷.

SVs contribute to phenotypic differentiation of heterotic groups

Two natural variants (SNP-1245 and Indel-2339) in the promoter regions of *ZCN8* were selected stepwisely during early domestication and northward expansion of maize²⁹. Interestingly, we found that all 14 FILs possessed the early-flowering haplotype in the *ZmCCT10* gene, consistent with the earlier report that *ZmCCT10* was domesticated before *ZmCCT9* during the post-domestication spread of maize³⁰. Among the 14 FILs, S37 retained the original late-flowering haplotype at *ZmCCT9*, *VGT1*, and *ZCN8*. Two temperate FILs (Chang7-2 and Huangzaosi) possessed the early flowering haplotype at SNP-1245 of *ZCN8*, but retained the late flowering haplotypes at *ZmCCT9*, *VGT1*, and Indel-2339 of *ZCN8*, whereas the other nine temperate FILs all possessed early flowering haplotypes at *ZmCCT9* and SNP-1245 of *ZCN8*. These observations suggest that *ZmCCT9*, *VGT1*, and *ZCN8* are important determinants of regional adaptation of tropical and temperate maize to their specific ecological niche. In line with this notion, we found that most tropical NAM founders²⁰ (12/13) possessed at least 3 late-flowering haplotypes (Supplementary Table 17), whereas all the temperate NAM founders possessed at least two derived, early-flowering haplotypes in *ZmCCT9*, *VGT1*, and *ZCN8* (Extended Data Fig. 4a-4c). PCR-genotyping of 350 temperate ILs¹ indicated that 96.2% of IDT germplasm possessed no less than 3 early-flowering haplotypes, whereas more than half of PA, TSPT, PB germplasm possessed no more than 2 early-flowering haplotypes (Extended Data Fig. 5). In general, the cumulative effect of early flowering haplotypes at *ZmCCT9*, *VGT1*, and *ZCN8* was consistent with the flowering phenotypes of these heterotic groups (Extended Data Fig. 6). Thus, *ZmCCT9*, *VGT1* and *ZCN8* likely contributed to the differentiation of flowering time of the various heterotic groups of temperate maize and should be the preferred targets for selection to improve the adaptability of PA, TSPT, PB germplasm in future temperate maize breeding.

Kernel row number (KRN) is a key component of maize yield. The female parents used in maize hybrid breeding, such as the SS heterotic group FILs, always have more KRN than the male parents (such as the NSS heterotic group)³¹. Previous studies have reported that a 1.2 kb transposon-containing insertion located ~60 Kb downstream of *UB3* is responsible for a major quantitative trait locus, *KRN4*³². Interestingly, we found that two SS FILs, B73 and A632, both harbored the haplotype in favor of higher KRN (Extended Data Fig. 4d). PCR genotyping of 350 temperate ILs¹ further indicated that 72.6% of SS germplasms harbor the haplotype in favor of higher KRN, which was much higher than other heterotic groups, hinting that the 1.2 kb transposon-containing insertion is likely a causal variation for higher KRN in the SS germplasm. In addition, as previously reported³¹, we found that OH43, Mo17 had longer ear lengths (EL) and larger kernel numbers per row (KNR), whereas Huangzaosi, Chang7-2, and Jing92 had relatively shorter EL and lower KNR (Extended Data Fig. 4e). Interestingly, we found that EL and KNR correlated well with the two haplotypes caused by insertion of Harbinger-like TE in the 5'-UTR of *KNR6*, which encodes a serine/threonine protein kinase that regulates maize grain yield³³ (Extended Data Fig. 4e). PCR genotyping of 350 temperate ILs¹ indicated that 83.9% of the NSS germplasm possessed the longer EL haplotype, while 50.0% of the TSPT germplasm possessed the shorter EL haplotype, hinting that this SV is likely a major causative variation underlying the differential EL and KNR characteristics of the NSS and TSPT groups (Extended Data Fig. 6).

Xu178 is a representative of the PB-group germplasm, which is derived from the US hybrid 78599, a cultivar known for superior resistance to maize rough dwarf disease (MRDD) (Extended Data Fig. 4f)¹². We conducted QTL mapping of MRDD resistance using a recombinant inbred line (RILs) population of Xu178 (MRDD resistant) and HuangC (MRDD susceptible, Extended Data Fig. 4f-4g). We detected a major QTL on chromosome 8 which overlaps with the previously reported *qMrdd1* locus³⁴. Interestingly, we found that among the 14 FILs, only Xu178 harbored the gain-of-function variant (caused by insertion of a *Helitron* TE) in the resistance gene *ZmGDIa*³⁴ (Extended Data Fig. 4h). PCR genotyping of 350 temperate ILs¹ further confirmed that the *Helitron* insertion was a variation specific to PB-group that underlies the resistance of Xu178 to MRDD (Extended Data Fig. 5). In addition, chromosome 6 of both Mo17 and OH43 contains a ~2.9 Mb large SV. This region contains the reported major SCMV resistance gene *ZmTrxh*³⁵. Thus, it is likely that the absence of this large SV underlies the susceptibility of Mo17 and OH43 to SCMV³⁶.

PCR genotyping of 350 temperate ILs¹ indicated that ~43.5% of the NSS and 57.1% of the X-group germplasm lack this large SV. In contrast, only 3.2% of the SS germplasm and 4.2% of the TSPT germplasm lack this large SV, hinting that this large SV is a major locus differentiating NSS and X-group from SS and TSPT germplasms.

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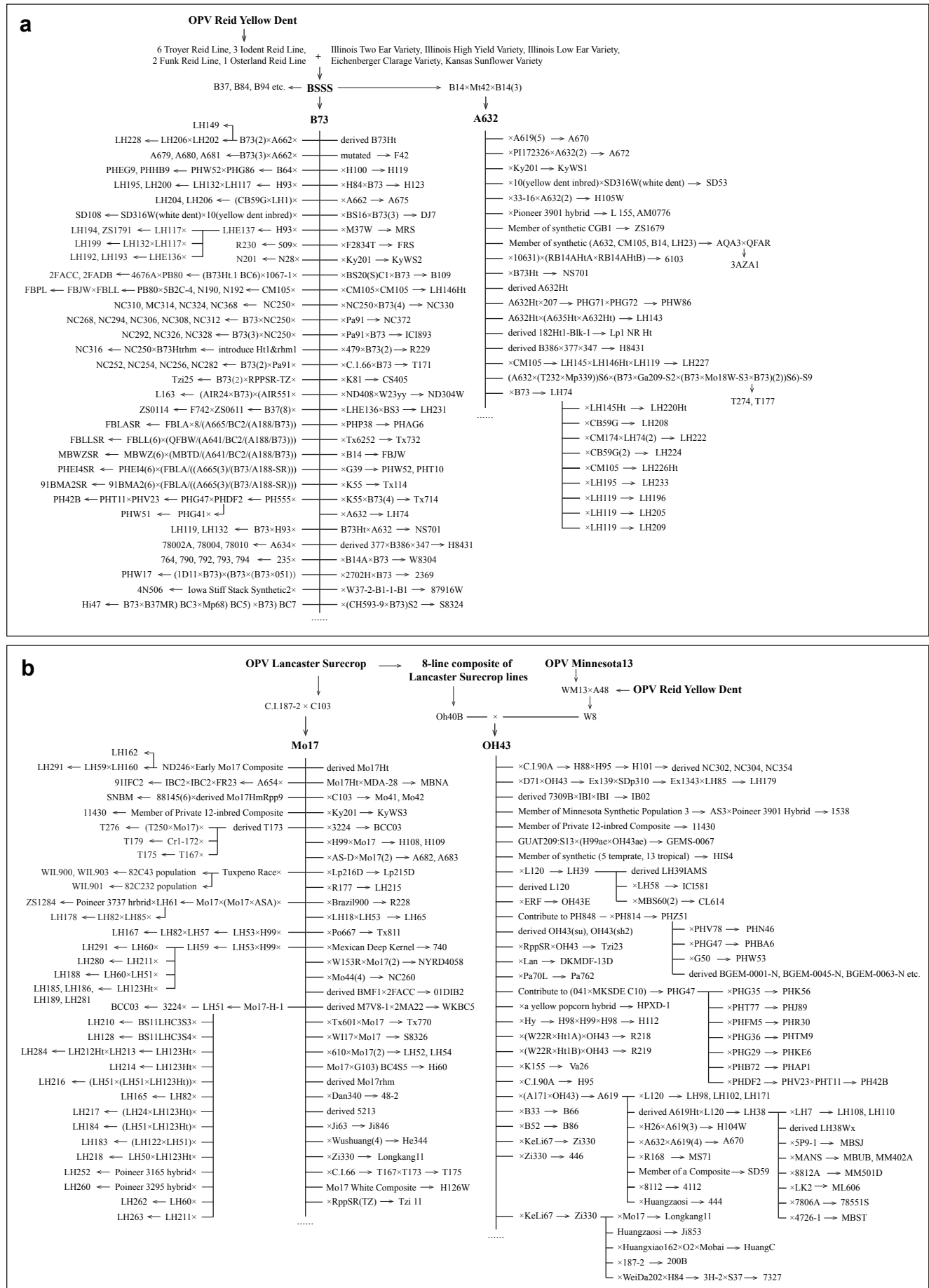
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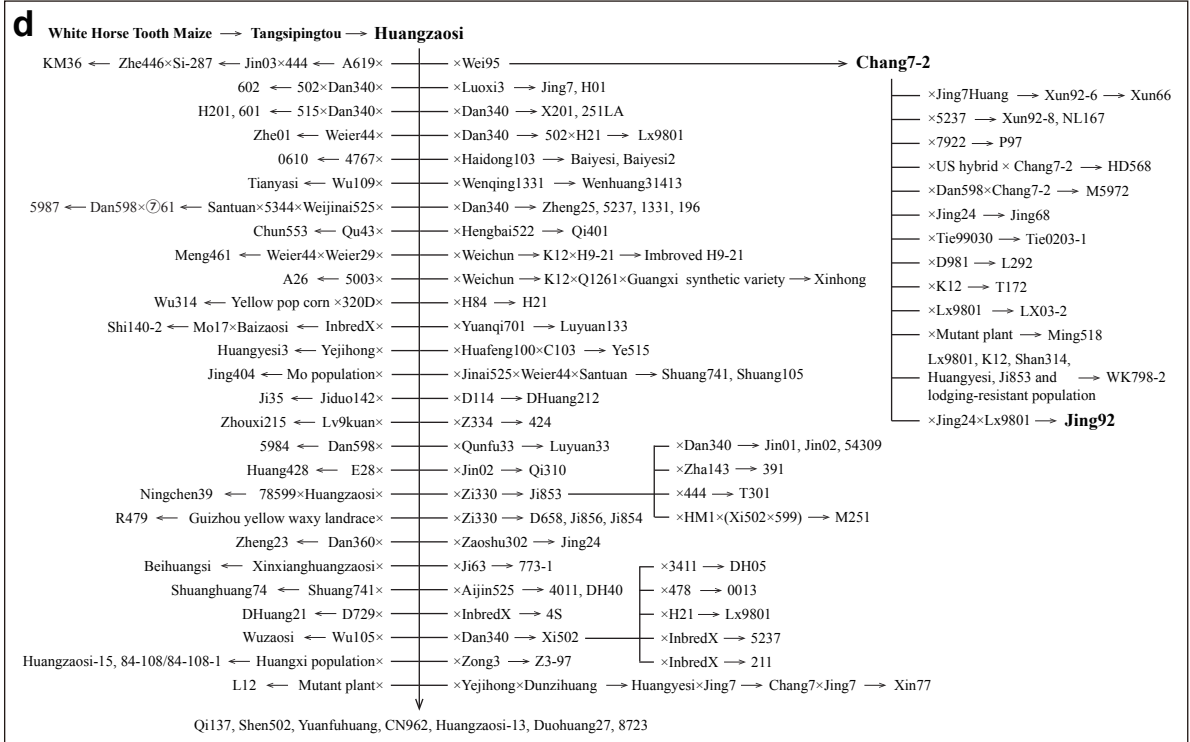
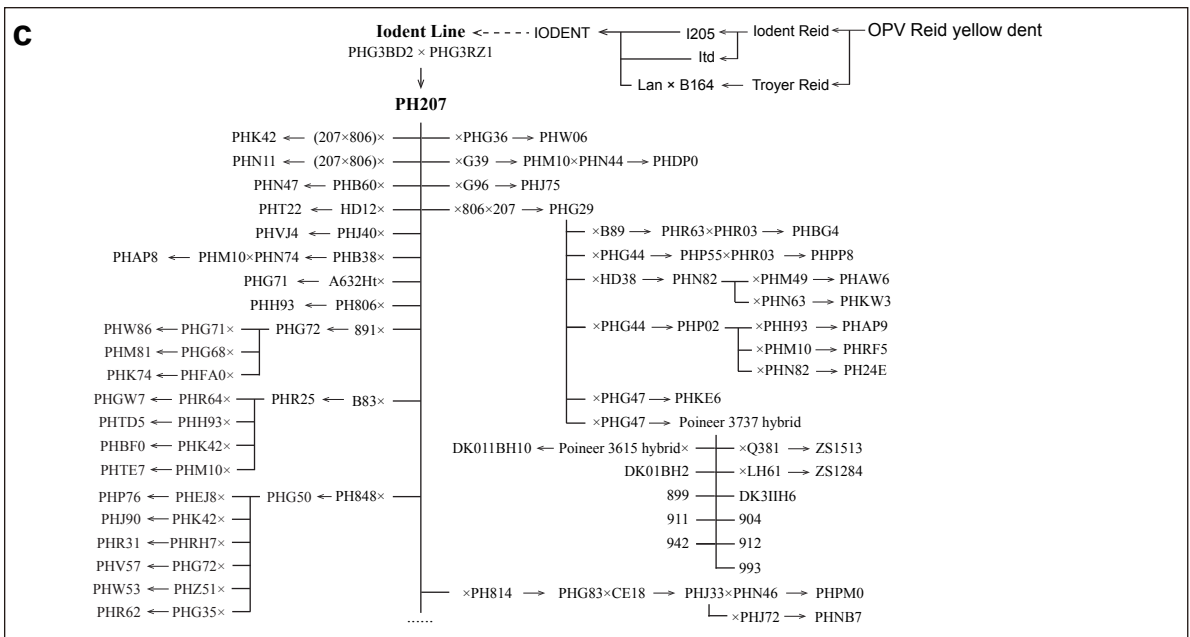
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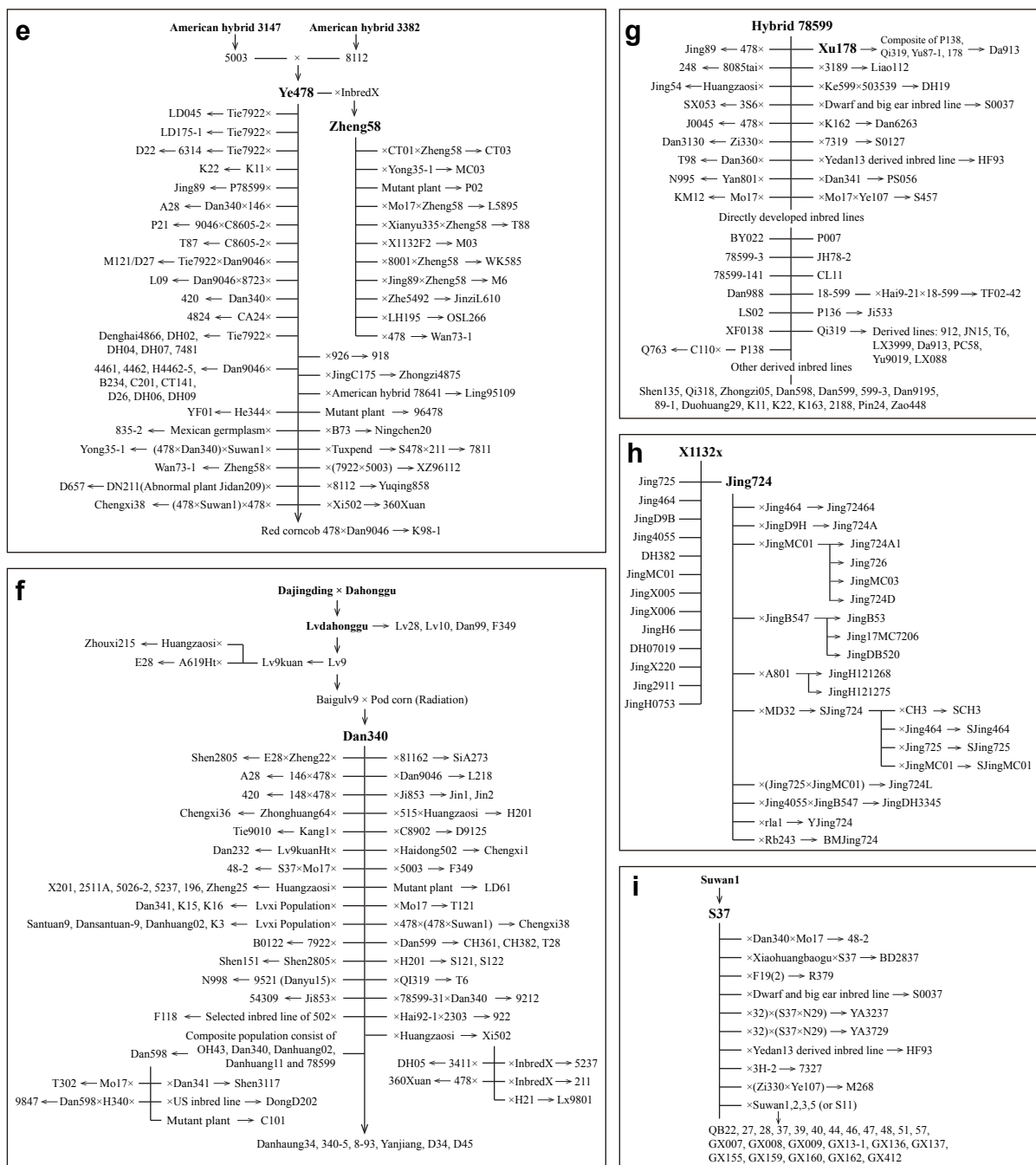
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Supplementary Figure 1







Supplementary Figure 1. Pedigree profile of the 14 maize FILs and related germplasm. The 14 FILs were used to develop dozens or hundreds of elite parental lines during modern hybrid maize breeding.